

The empirical recurrence for OEIS A303515, recovered from its tail

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Abstract

OEIS A303515 records “Empirical recurrence of order 63” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 68$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 3$, so the recurrence is proved for every $n \geq 66$. Its last coefficient is $c_{63} = 512 \neq 0$, so no further tail satisfies anything shorter; any order-63 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A303515 is “Number of $4 \times n$ 0..1 arrays with every element equal to 0, 1, 2, 4, 5 or 6 horizontally, diagonally or antidiagonally adjacent elements, with upper left element zero.”. It has offset 1 and begins

8, 81, 420, 1596, 7951, 43448, 222205, 1111647, ...

The entry states:

Empirical recurrence of order 63 (see link above)

As of the “Last modified” line on the live entry (Apr 25 2018 12:30:30, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 68$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 68$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 63: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 3$ is the first whose minimal order is exactly the stated 63.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 136$ exact terms from $n = 3$ on, it returns order 63 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{63} c_i a(n-i),$$

c_1	16	c_{17}	−14183312	c_{33}	22627267	c_{49}	6519792
c_2	−84	c_{18}	24060891	c_{34}	182878910	c_{50}	5987478
c_3	58	c_{19}	−11616803	c_{35}	−49201135	c_{51}	4510104
c_4	1008	c_{20}	7458158	c_{36}	136693571	c_{52}	6249624
c_5	−3746	c_{21}	33898855	c_{37}	−53060036	c_{53}	1760040
c_6	3674	c_{22}	−85855381	c_{38}	−100171224	c_{54}	−3231168
c_7	12885	c_{23}	49914668	c_{39}	3944625	c_{55}	−2581368
c_8	−74557	c_{24}	−28446628	c_{40}	−114972922	c_{56}	−31144
c_9	143814	c_{25}	−69651250	c_{41}	−18414095	c_{57}	414768
c_{10}	157623	c_{26}	209343427	c_{42}	41358381	c_{58}	49904
c_{11}	−1027792	c_{27}	−93118661	c_{43}	47642675	c_{59}	82880
c_{12}	925233	c_{28}	56643184	c_{44}	59010814	c_{60}	58496
c_{13}	1260579	c_{29}	38837222	c_{45}	15512366	c_{61}	−19456
c_{14}	−3436635	c_{30}	−289292227	c_{46}	−15344793	c_{62}	−9728
c_{15}	5895434	c_{31}	116981114	c_{47}	−35480756	c_{63}	512
c_{16}	−3788216	c_{32}	−68154724	c_{48}	−19416208		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 66$, and it is the recurrence OEIS A303515 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 3$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{63} - c_1 x^{62} - \dots - c_{63}$. Its constant term is $-c_{63} = -512$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 63 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 63, so $\deg q = 63$, and it is monic. It holds from some index t on. From $\max(t, 3)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 63, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 20 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 63, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 512.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-63 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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